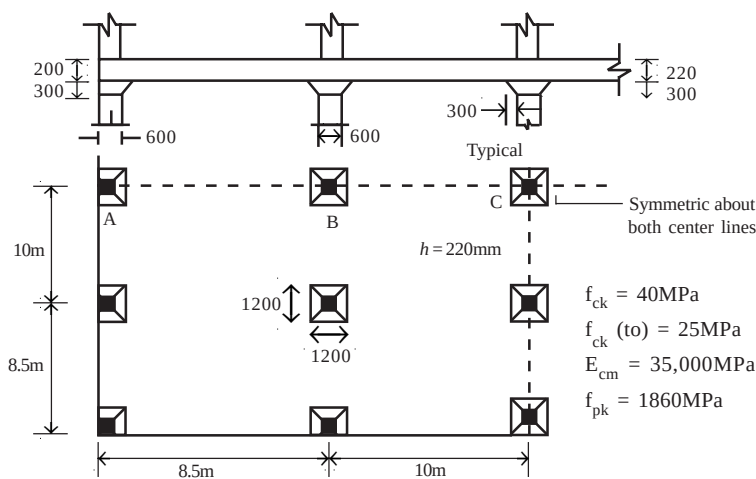


CE-8003(1) (CBGS)
B.E. VIII Semester Examination, June 2020
Choice Based Grading System (CBGS)
Pre-Stress Concrete Design
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
 ii) All questions carry equal marks.
 iii) Draw neat diagram.

1. a) What is Prestressed concrete? Explain the concept of prestressing. 7
 b) Explain Load Balancing Concept with neat sketch. 7
2. a) A prestressed concrete beam of 150mm×300mm cross-section supports a live load of 5 kN/m over a simple span of 8m. It has a parabolic cable having an eccentricity of 75mm at the midspan and zero at the ends. Determine the force of prestress if the net resultant stress at the bottom fibre at midspan is zero under the action of dead load, live load and prestress force. 7
 b) A post-tensioned bonded beam of rectangular section 200mm wide and 400mm deep has an effective cover of 60mm. Determine the flexural strength of the section if it is prestressed with 30-5 mm high tensile steel wires. Assume $\sigma_{ck} = 50\text{N/mm}^2$ and $\sigma_p = 1600\text{ N/mm}^2$. 7
3. Determine the tendon required in the 220mm thick flat slab shown in the following Fig. The live load of the slab is 3KPa and Deadload is 1 Kpa (+ self weight). All columns (600×600) mm and 4m long above and below the slab. At the top of each column a 300mm column capital is used to increase the supported area. 14



[2]

4. Design a pretensioned prestressed rectangular section if it is required to support a dead load of moment of 15 kN/m (including self weight), and a live load moment of 30 kN/m. Use M40 concrete and suitable data. 14
5. Write down the general design aspects of pretensioned pre-stressed bridge deck. 14
6. a) Write down the type of prestressing. Also define the losses in pre-tensioning. 7
b) A simply supported unbonded post tensioned beam of 300mm×500mm cross-section has an effective span of 7m. It is prestressed with having a total cross-sectional area of 845 mm² at an eccentricity of 150mm. Determine the moment of resistance if $\sigma_{ck} = 40$ MPa, $\sigma_p = 1800$ MPa and effective prestress $\sigma_{pe} = 0.5 \sigma_p$. 7
7. The end block of a post tensioned PSC beam 300mm wide and 300mm deep is subjected to a concentric anchorage. Force of 800 kN by a freyssinet anchorage system of area 11000 mm². Design and detail the anchorage reinforcement for the end block. 14
8. Write short notes on: 14
 - i) End Block
 - ii) Partial Prestressing
 - iii) Creep Loss
 - iv) Stresses induced in concrete during circular prestressing.
